

MACROECONOMETRIE

2026-2027



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- GitHub: https://github.com/vgautier1/cours_M2_IEF

- 8 * 3.5-hour sessions co-taught with Florian Pothin
- 4 macroeconomic concepts, 7 hours per concept
- Exam:
 - Project to submit at the end of each concept (report + data + code/script)
 - 2 out of 4 projects graded, or average of the 4 projects (to be confirmed)
- Report content:
 - 1–3 pages on theory + introduction of the research question + stylized facts + outline
 - 2–5 pages on the econometric application (explanation of methodology and results + tests)
 - 1–2 pages of conclusion (link between theory and econometric results)

Be careful with the use of generative AI

PHILLIPS CURVE – REVIEW

- Illustrates a negative empirical relationship between unemployment rate and inflation rate
- Central concept in macroeconomics, especially for economic policymaking
- Challenged during the 1970s
- Modern Phillips curve and limitations

I- Origins (1950s-1960s)

- A.W. Phillips (1958) studied the macroeconomic situation of the UK between 1861 and 1913 and found an inverse relationship between the unemployment rate and the rate of change of nominal wages.
- When unemployment is high, nominal wages increase slowly because workers are not in a position to bargain due to the large available labor force. The opposite effect is observed when unemployment is low: workers are in a strong bargaining position due to scarce available labor and can negotiate nominal wage increases.

⇒ ***Negative relationship between unemployment and wage inflation***

- Neoclassicals: wages are linked to the productivity of labor

⇒ ***The Phillips curve contradicts neoclassical theory***

- Keynesians: active economic policies, such as stimulating aggregate demand, can influence unemployment and inflation.

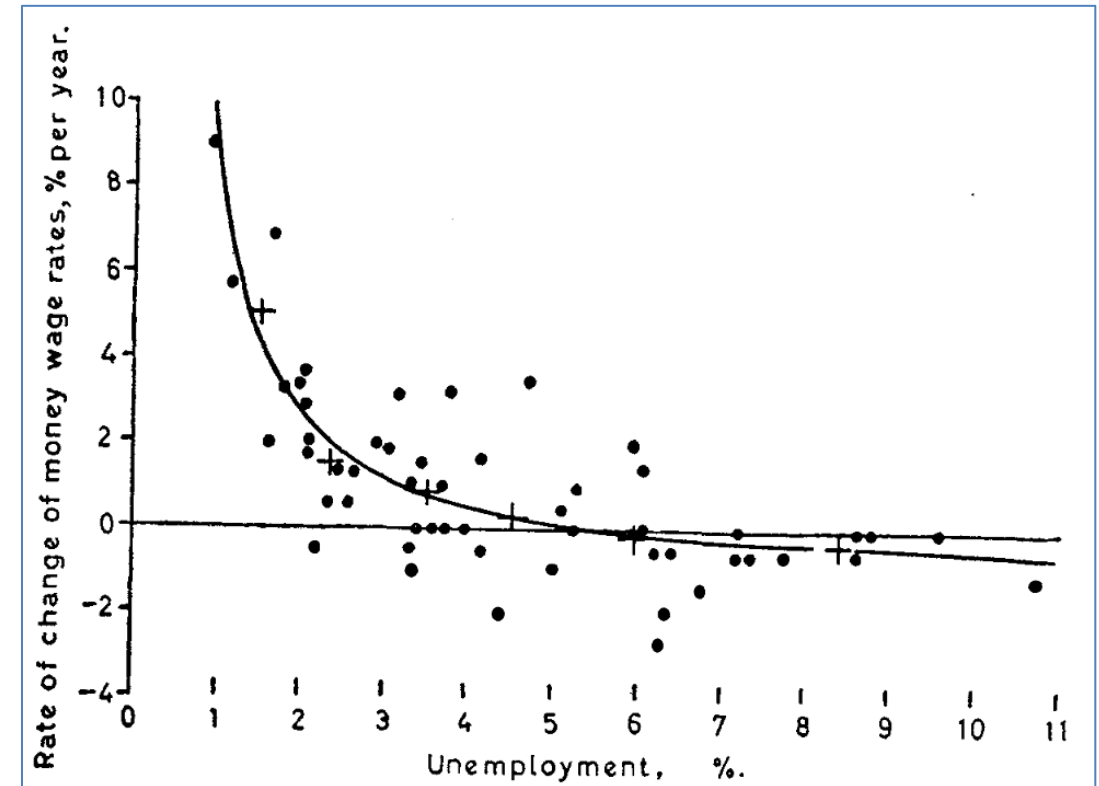
⇒ ***The Phillips curve supports Keynesian thought***

I- Origins (1950s-1960s)

- The original Phillips curve describes a **negative** and **non-linear** relationship between the unemployment rate and the growth rate of nominal wages.
- **Empirical** relationship, as it is based on observations.

$$\frac{\Delta W}{W} = \beta U \quad (1)$$

W nominal wages et U unemployment rate.



Source: Phillips (1958).

- Theorization of the Phillips curve by R. Solow (Keynesian) and P. Samuelson (Neoclassical) when focusing on the US.
- => They extended Phillips' relationship to inflation because wage increases are passed on to prices (via production costs) and to demand (gains in purchasing power).**
- **Integration into the Keynesian model:**
 - Unemployment: labor market disequilibrium caused by insufficient aggregate demand.
 - Phillips Trade-off: Stimulating aggregate demand reduces unemployment (via higher output) but accelerates inflation due to wage and price pressures.
- => Policymakers must choose an 'optimal' equilibrium point allowing moderate inflation and unemployment according to their preferences, or accept higher inflation to boost demand/reduce unemployment.**

II- Challenging the Theory (1970s) – Phillips Curve Horizon

- Critique by M. Friedman and E. Phelps: the **inverse relationship between unemployment and inflation disappears in the long run**. Their argument is based on the fact that economic agents (consumers, firms) adapt their inflation expectations based on past conditions, a concept known as **adaptive expectations**.
- Gives rise to the expectations-augmented Phillips curve:

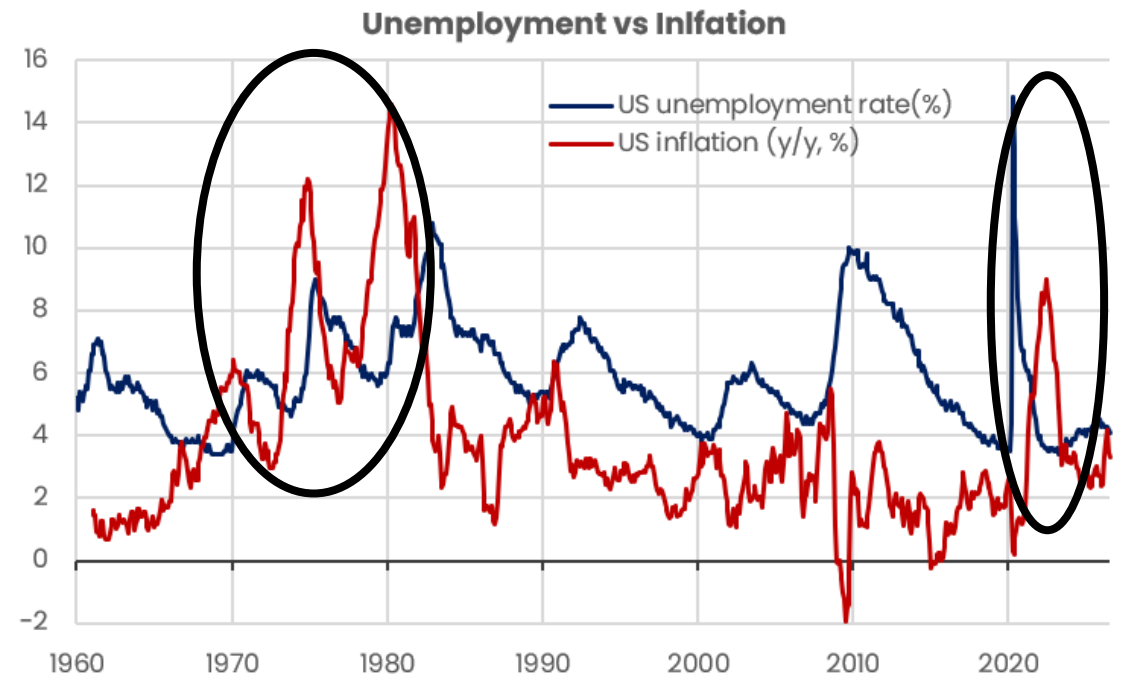
$$\pi_t = \pi_t^e - \alpha(U_t - U_t^N) \quad (2)$$

π_t^e expected inflation and U_t^N the natural rate of unemployment (NAIRU).

- Friedman and Phelps introduced the concept of NAIRU, representing a level of unemployment compatible with stable inflation. At this level, there is no longer a trade-off between unemployment and inflation. Any attempt to reduce unemployment below this level through demand stimulus policies simply leads to an acceleration of inflation.
- **The trade-off between unemployment and inflation highlighted by the traditional Phillips curve is only valid in the short run. In the long run, expansionary policies aimed at maintaining artificially low unemployment only increase inflation without any lasting effect on unemployment (vertical long-run Phillips curve).**

II- Challenging the Theory (1970s) – Accounting for Supply Shocks

- The stagflation (coexistence of high unemployment and high inflation) of the 1970s challenged the foundations of the traditional Phillips curve.
 - 1970s: Major supply shocks, notably oil shocks, increased production costs and reduced output, raising unemployment.
- ⇒ ***The traditional Phillips curve does not hold in the presence of a supply shock ⇔ imitation of an exclusively demand-based approach to understanding inflation.***



III – New Keynesian Phillips Curve

- Evolution from the Keynesian model to the New Keynesian model: nominal rigidities explain why economic policies can have a short-term effect on the real economy, it is no longer surprises (adaptive \$ \to \$ rational expectations) + microfoundations. Gives rise to the **New-Keynesian Phillips Curve** (NKPC):

$$\pi_t = \beta E_t(\pi_{t+1}) + K(y_t - y^*) \quad (3)$$

$E_t(\pi_{t+1})$ the rational expectation of future inflation (past inflation, immediately available information, and future beliefs), K the slope of the PC and $y_t - y^$ the output gap (real vs. potential output). An alternative version exists with marginal production costs instead of the OG. potentielle).*

⇒ **Remains a demand-based model, with no integration of supply, so it does not explain the stagflation of the 1970s.**

$$\text{Extended NKPC : } \pi_t = \beta E_t(\pi_{t+1}) + K(y_t - y^*) + \varepsilon_t \quad (4)$$

ε_t an exogenous shock (supply shock on production costs, fiscal shock, or sector-specific shock).

- LaThe NKPC fits into DSGE, which is the New Keynesian model (IS / Taylor Rule / NKPC).

IV – Validity and Relevance of the Phillips Curve

- Current debates focus on the validity and relevance of this relationship in the contemporary economy, particularly in the context of inflation, monetary policy, and global economic conditions.
- Prior to Covid, many economies experienced historically low unemployment rates without facing high inflation.
- Inflation remained relatively stable despite fluctuations in unemployment.

⇒ ***Flattening of the Phillips curve***

- Recent events (Covid crisis and war in Ukraine) show that supply shocks are still poorly captured by the Phillips curve. The pandemic led to supply disruptions, price spikes, and labor shortages in certain sectors, causing inflationary pressures despite relatively high unemployment rates in some regions. The war in Ukraine also led to increases in energy and commodity prices, exacerbating inflationary tensions.

IV – Validity and Relevance of the Phillips Curve

- **More anchored inflation expectations / more credible central banks (inflation targeting framework):** Anchoring inflation expectations around central bank targets (e.g., 2% inflation) can make the Phillips curve less sensitive to unemployment. When agents expect monetary authorities to act swiftly to return inflation to target, the effect of the unemployment gap on inflation becomes less pronounced.
- **Diversification in supply chains through globalization:** Importing cheaper goods, particularly from low-labor-cost countries, has limited inflationary pressures in advanced economies by reducing their production costs.
- **Technology and innovation:** Technological innovation and digitalization have also had a deflationary effect, by reducing production costs and enabling productivity gains.
- **More structural unemployment:** Increase in the structural component of unemployment due to technological progress (demand for more technical/specialized skills), globalization (loss of industrial jobs in mature economies), demographic shifts (aging population), and stricter regulations (rigid contracts and high firing costs preventing quick labor market adjustment).

PHILLIPS CURVE – CASE STUDY

1) Phillips Curve in the Euro Area – Analysis of the impact by unemployment type

- Compare the Phillips curve in two Euro Area economies, one with predominantly structural unemployment and the other cyclical

2) Phillips Curve in the Euro Area – Analysis of the impact by energy dependence

- Compare the Phillips curve in two Euro Area economies, one heavily dependent on energy imports (especially oil) and the other not

3) Phillips Curve in the Euro Area – Evolution of the slope

- Analyze the Phillips curve over different characteristic sub-periods of the Euro Area (inspired by BdF 2018 paper)

4) Phillips Curve worldwide – Mature VS Emerging/Developing Economies

- Compare the Phillips curve in two global economies, one mature and one emerging (focusing on inflation expectation anchoring)

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- Phillips curve to estimate according to topics and project progress:
 - Inflation ~ unemployment
 - Inflation ~ adaptive expectations + unemployment
 - Inflation ~ adaptive expectations + unemployment + oil (or other energy)
 - Inflation ~ output gap
 - Inflation ~ adaptive expectations + output gap
 - Inflation ~ adaptive expectations + output gap + pétrole (ou autre énergie)

Attention à l'utilisation de l'IA générative

- **Unit root:** When time series are stationary, relationships between variables (e.g., dependency between dependent and independent variables) remain constant over time. This provides reliable parameter estimates. If series are non-stationary, relationships can change over time, making estimates unstable and potentially biased.
 - Augmented Dickey-Fuller test (ADF):
 - H_0 : Non-stationarity
 - H_1 : Stationarity
- **No residual autocorrelation:** When residuals are autocorrelated (i.e. dependent on each other), it suggests the model does not capture all information in the data and systematic unexplained patterns remain. This can lead to inefficient coefficient estimates with incorrect standard errors.
 - Durbin-Watson test:
 - H_0 : No autocorrelation
 - H_1 : Autocorrelation

- **Homoscedasticity of residuals:** Homoscedasticity guarantees that regression coefficient estimates are efficient. If residual variance varies (heteroscedasticity), estimates can be inefficient or standard errors distorted.
 - Breusch-Pagan test:
 - H_0 : Homoscedasticity
 - H_1 : Heteroscedasticity
- **Normality of residuals:** Significance tests (t-tests, F-tests) rely on the assumption that residuals follow a normal distribution. If violated, p-values may be misleading, potentially leading to incorrect conclusions on variable significance.
 - Jarque-Bera test
 - H_0 : Normal distribution
 - H_1 : Non-normal distribution

Contacts

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